Daniel Guo

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Education

University Of Waterloo — Waterloo, ON
Scientific Computing and Machine Learning - Applied Mathematics — *Expected 2030*
Presidential Scholar's Award, Provincial (BC) Top Achievement Scholarship

Experience

Research Assistant — Aug 2026 – Dec 2026
SFU Intelligent Robotics Lab — *Vancouver, British Columbia*

- Engineered CUDA-backed **3D vision pipeline** integrating VLMs and point-cloud segmentation to recover articulated object joints under Dr. Siqi Zhou for multi-point articulation modelling.
- Accelerated processing pipeline from **24.4s to 0.5s (48x speedup)** by migrating CPU post-processing tasks to parallel GPU workloads.
- Built **point-ID render maps** that trace VLM-selected image pixels back to the original 3D scan, then retain those points when Point-SAM reduces a 67K-point scan for GPU inference.

Software Engineer Intern — May 2026 - Aug 2026
ZigZag Technologies Inc. — *Toronto, Ontario (Remote)*

- Built and tested control layer for autonomous vending machines, with **CLI and REST API** for autonomous vending machines + runtime adapter that maps simulated operations to live machine APIs, for controlled rollout.
- Integrated with an external backend through a typed adapter that validated responses, mapped network and service failures, and isolated unsupported operations.
- Added persistent, **idempotent vend-command handling**: a request is recorded before dispatch and must be reconciled after a lost response instead of being sent to the machine again.

Software Developer Intern — April 2026 - May 2026
Legends Haul — *Vancouver, British Columbia*

- Implemented two stage signature detection algorithm for automating AR data entry: OpenCV contour-based heuristics + **finetuned CPU optimized YOLO V8-based model** with hard negative mining training loop to distinguish noisy signatures vs handwritten notes on forms at **94% accuracy** (from 82% in **just 15 epochs**)
- Automated manual AR data entry processes for shipping logistics, **saving up to 4 hours of work per day**, and **rectifying over 500 past data entry errors** with OCR based image detection with Excel records cross-checking

Intern — June 2025 - Aug 2025
Sharpcone — *Langley, British Columbia*

- Worked on quote drafting tool for plastic supplier to generate customer-ready PDFs for draft quotes (incorporating pre-existing Excel-based pricing data), **saving up to 5 hours weekly**

Projects

Phone Detection Roomba | *Python, OpenCV, ESP32, AWS, YOLO-V8*

- Engineered a CV pipeline: an on-device **YOLOv8 detector** for frame-rate phone/face tracking for minimizing network latency, plus a cloud AWS Rekognition face-recognition call to identify the person
- Implemented the patrol controller as a finite state machine using **proportional visual-servoing control** to steer the Roomba toward a target via bounding-box position.

ASCII Fluid Dynamics Simulator | *Javascript, HTML Canvas, CSS*

- Implemented Jos Stam's 1999 Stable Fluids paper's fluid simulation for my personal website danielguo.net
- Tracked density and velocity via **semi-lagrangian scheme** injected ASCII based on velocity and density

VR 3D Modelling App | *C#, Meta XR SDK, OpenXR, Unity*

- Developed with team for hackathon at Carnegie Mellon, **won Honorable Mention (out of 500 submissions)**

Skills

Languages: Python, C/C++, C#, TypeScript/JavaScript, Java, SQL, Bash
Full Stack & Cloud: React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, Tailwind
ML/Robotics: PyTorch, TensorFlow, OpenCV, YOLO, CUDA, ROS 2, ESP32
Tools/Platforms: Git, GitHub Actions, Docker, Linux, REST/GraphQL APIs, AWS (Lambda, S3, SQS)